



Agile governance frameworks for strengthening public financial management in emerging economies

Sodiq Ogunmola^{1*}

¹4400 S M L King Jr Pkwy, Beaumont, TX 77705

*Corresponding author: sogunmola@lamar.edu

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Abstract

Public Financial Management (PFM) systems are still afflicted in many emerging economies by some of their timeless challenges: bureaucratic rigidity; limited fiscal transparency; archaic or obsolete legacy systems; weak inter-agency coordination; and slowness to respond to economic shocks. All these factors undermine government capacity to plan, allocate and monitor public resources along a time and with accountability. Recently, agile governance rooted in approachability, adaptable iterative cycles for decision-making and collaborative yet largely data-informed structures has surfaced as an exciting approach to usher the public sector into the modern world with its long traditions of procedural inflexibility. This article studies how agile governance frameworks could strengthen Public Financial Management (PFM) in emerging economies by enhancing fiscal responsiveness, improving efficiency in budget execution, promoting transparency and citizen engagement. It synthesizes the extant literature from organizations such as the OECD, IMF, and World Bank with other reputable academic literature and situates agile governance within theoretical frameworks such as Complexity Theory, Institutional Theory and Public Value Governance. Country-specific illustrations include Nigeria, Ghana, Kenya, India, USA, and UK, showing where and how agile approaches could be applied to budgeting, procurement, auditing, treasury operations and digital PFM reforms. The article advances the Agile PFM Model and Digital PFM Architecture, which encompasses modular digital systems, interoperable data layers, regulatory sandboxes, rapid feedback loops, performance dashboards, and cross-functional PFM teams. The findings suggest that agile governance, anchored on political will, digital infrastructure, and institutional safeguards, could significantly boost fiscal resilience, reduce leakages, and allow governments to respond more rapidly to other emergencies, including pandemic, economic instabilities, and natural disasters. The paper elucidates an implementation roadmap together with risk mitigation considerations for the attention of policymakers, donors, and PFM modernization teams

Keywords: Agile Governance; Public Financial Management (PFM); Emerging Economies; Fiscal Transparency; Adaptive Governance; Digital PFM Systems; Agile Budgeting; Agile Procurement; Regulatory Sandboxes; Institutional Reform; Public Sector Innovation; Complexity Theory; Fiscal Resilience; Public Value Governance; Government Accountability; Performance-Based Budgeting; Interoperability; Treasury Modernization; Digital Transformation; Public Administration Reform.

Introduction

Public Financial Management is the institutional base through which governments organize all these matters relating to management of public resources, encourage fiscal discipline on the one hand, and deliver essential services to citizens on the other. Unfortunately, many developing countries, with respect to public financial management systems, often come with structural flaws-inflexible budget structure, legacy systems that have been fragmented, the regulatory framework is not up to date and real-time financial data is limited; making them less transparent and responsive to fiscal matters (World Bank, 2019; IMF, 2020). Thus, these systemic limitations inhibit the collection and control of revenues, spending response to shocks, and the countries' growing public expectations of accountability. With change around the globe-coming from digitalization, increased volatility of economies, and greater demands from societies regarding transparency-modernization of PFM systems has become one of the most critical and urgent priorities (OECD, 2021).

Agile governance-this describes an approach that is adaptive to iterative cycles, fast learning, collaboration and citizen-centered design-stands well positioned to remedy these features, which have long plagued these systems. Furthermore, it encourages governments to experiment and make changes in real time based on information from ongoing learnings rather than forward-bounded linear planned processes, standard operating procedures, as in conventional bureaucracy (Mergel et al., 2021). It is indeed interesting to note that though agile methods were originally developed in software development, they have now found extensive application in the public administration sector as governments increasingly learn how to use agile methods to improve responsiveness and cut implementation bottlenecks (Luna et al., 2020).

Hence, countries such as Nigeria, Ghana, Kenya, India, the United States, and the United Kingdom adopting reforms in line with agile principles demonstrate the relevance of agile governance to PFM. For example, Nigeria's Treasury Single Account (TSA) has improved cash management and transparency with a single digitally unified platform (Adewuyi, 2022). Kenya's e-payment and e-procurement reforms have significantly reduced leakages and improved auditability (World Bank, 2020). India's Public Financial Management System (PFMS) holds real promise from real-time dashboards for improving expenditure tracking and speeding up fund disbursement (Mukherjee & Sikdar, 2020). Upon that, the Government Digital Service of the United Kingdom is often cited for its agile public procurement alongside its digital-first governance initiatives (U.K. Cabinet Office, 2019). The United States has also included some iteration of agile practices in its federal IT modernization through the U.S. Digital Service and reform of budget processes (GAO, 2021).

With economic shocks, political transitions, pandemics, and climate-related emergencies requiring rapid and coordinated fiscal responses, there is an increasing realization that the traditional format of PFM processes, most often annual, linear, and compliance-focused, is inadequate (IMF, 2021). Agile governance provides instruments that help build resilience by creating windows for timely adjustments to budgets, encouraging cross-functional collaboration, strengthening continuous feedback loops, and improving data-informed decision-making (OECD, 2020).

This paper contends that an embedding of agile principles in PFM systems can help enhance the public sector adaptability, fiscal transparency and institutional learning in a more significant way. Proposed is an integrated Agile PFM Model that embeds iterative budgeting, cross-functional teams, modular digital systems, regulatory sandboxes, and performance dashboards. The paper also proposes a Digital PFM Architecture to

support real-time analytics; enhance interoperability; and increase performance through better Treasury operations.

Literature Review

In over three decades, public financial management has garnished interest towards the reform agenda of emerging economies. Modern approaches to public finance management should enable governments to plan, allocate, monitor and manage public resources properly to strengthen fiscal disciplines and public accountability (Cangiano et al., 2013). According to Allen and Tommasi (2001) and the World Bank (2019), most of these emerging economies have serious structural problems-that is, the budget processes are fragmented, treasury systems are obsolete, internal controls are weak, and performance information is poorly integrated into budgets and reporting. Consequently, these structural weaknesses manifest inefficiency, delayed service delivery, and leakage, thus calling for innovations in public finance governance.

Traditional PFM Reform Literature

Traditional public financial management reform frameworks promulgated by international institutions such as the World Bank, IMF and OECD are meant to conceptually increase transparency, accountability and efficiency. Some of the major approaches are:

-Medium-term expenditure framework (MTEFs): MTEFs create multi-year perspectives on budget planning and implementation, encouraging governments to tie resources to policy priorities (Diamond and Khemani, 2006). MTEFs are more of a planning instrument. They are often inflexible and not well-suited to adapt to unforeseen fiscal shocks (IMF, 2020).

-Performance-based budgeting: (PBB) allocates to outputs, thus encouraging economy and results-oriented management (Shah, 2007). However, despite such a promise, the main bottleneck in most emerging economies in the implementation of performance budgeting continues to be poor implementation due to inadequate data infrastructure and capacity gaps (Alesina et al., 2008).

-Treasury reforms and Integrated Financial Management Information Systems (IFMIS): The adoption of IFMIS has resulted in real-time monitoring of expenditures and greatly enhanced financial reporting. The studies reveal that if implemented, it enhances transparency and reduces risks of corruption, but even then, one of the most persistent challenges is system interoperability (World Bank, 2020; Mukherjee & Sikdar, 2020). While all these reforms contribute to strengthening public finance management, scholars have noted that their linear and compliance-oriented structures impede responsiveness and adaptability (Cangiano et al., 2013; Allen and Tommasi, 2001). Some new issues emerging today, like rapid urbanization, pandemics, and fiscal shocks, are continuously demanding more flexible and iterative public finance management approaches.

Digital PFM and Adaptive Systems

Digital transformation accommodates the enabling function of agile public financial management. For instance, with the establishment of digital platforms like treasury systems, e-procurement portals, or integrated financial dashboards, governments can adjust the fiscal data on an almost real-time basis, monitor expenditure execution, and adjust allocations in a fast manner (Long et al., 2023). For example:

- In Nigeria, TSA and GIFMIS allow for improved cash management by centralizing government accounts for

real-time expenditure monitoring (Adewuyi, 2022).

- **PFMS in India** has allowed for direct benefit transfers and instant fund tracking; this shows how digital systems enable rapid adjustment and hold the fiscal system accountable (Mukherjee & Sikdar, 2020).

- **Kenya's eCitizen and IFMIS integration** indicates how digital platforms encourage iterative fiscal decision-making, as well as foster transparency (World Bank, 2020).

Literature reveals that digitization improves coastal effectiveness as well as lays the foundation for dynamic budgeting, procurement and performance monitoring (OECD, 2021). Digital systems coupled with adaptive governance will give the government ample room to iterate, pilot-test and respond to various shocks in addition to self-regulation.

Agile Governance in the Public Sector

Originally introduced concerning software development, agile governance is now increasingly adopted in the field of public sector administration due to the iterative decision-making, cross-functional collaboration as well as learning-in processes that it espouses (Mergel et al., 2021; Luna et al., 2020). The principles include:

-**Iterative Planning-Execution:** Structured for short cycles of planning, implementation, and evaluation, it allows a quick real-time simulation of possible changes such as modifying decisions (OECD 2020).

-**Cross-Functional Teams:** Encourages the collaboration of finance, procurement and sectoral experts through collaborative teams, thus enhancing coordination and reducing bottlenecks (Mergel et al., 2021).

Regulatory Sandboxes and Pilot Programs: Allowing controlled experimentations whereby governments can trial fiscal mechanisms or digital platforms without risking the entire system. (OECD, 2024).

Data-Driven Decision-Making: Regularly tracking key performance indicators helps ensure that any changes to policy or budgeting are grounded in solid evidence (Long et al., 2023).

Although research in this area is still developing, examples from the UK, the United States, and India show that agile governance can improve how quickly governments respond, reduce inefficiencies, and strengthen public trust (U.K. Cabinet Office, 2019; GAO, 2021; Mukherjee & Sikdar, 2020).

Integrating Agile Governance with PFM

Research shows that making public finance management (PFM) systems more agile requires both meaningful institutional reforms and reliable digital infrastructure (OECD, 2021; Mergel et al., 2021).

-**Performance and Risk Dashboards:** Data on actual fiscal performance: This instrument is expected to help decision-makers course through as circumstances change (World Bank, 2019).

-**Digital Procurement and Treasury Platforms:** Automated systems simplified the processes, reduced time delays, and further enhanced transparent operations on finances (Long et al., 2023).

-Skills and Capacity Development: Public servants are trained and further developed with opportunities for constant learning to build their confidence in agile methods in their work (Adewuyi, 2022). This gap underscores the need for studies like this paper, which proposes a holistic Agile PFM Model and illustrates practical applications in emerging economies such as Nigeria, Ghana, Kenya, and India.

Theoretical Framework

The theoretical framework bears the focus upon how an agile governance model could reinforce Public Financial Management (PFM) in emerging economies. It sets the ground basis for understanding how multiple theories can bring together a multidimensional analysis of dynamic, institutional, and value-based aspects of fiscal governance. This paper draws on Complexity Theory, Institutional Theory, and Public Value Governance to explain why reforms are needed and how it can guide the design of an Agile PFM Model.

Complexity Theory

Complexity theory posits that social settings, including public sector institutions, function as complex adaptive systems, whereby components are interdependent, relationships are nonlinear, and behaviors are emergent (Holland 1995; Snowden & Boone 2007). The public financial management (PFM) system presents a rapidly transforming economic, political, and social landscape for governments, imposing shocks and uncertainty on linear planning that is not equipped to address such rigidities (OECD 2020). The following stems from applying Complexity Theory to PFM: that the policy must be subject to iterative decision-making because fiscal policies and the budget must respond to the unpredictable components itself, for instance, commodity price changes because of a pandemic and natural disasters (Mergel et al., 2021).

Moreover, interdepartmental and cross-functional interactions among the finance ministries, sector-agencies, and regulators are required for emergent problem-solving and adaptive responses (Long et al., 2023). Finally, feedback loops of real-time financial data and performance monitoring allow governments to iteratively improve their strategies for implementing more responsive fiscal management (Mukherjee & Sikdar, 2020). In effect, Complexity Theory provides the foundation for contextualizing the agile principles of iterative planning, quick adaptation, and systemic learning within PFM.

Institutional Theory

Institutional Theory posits that formal rules, procedures, norms, and culture influence organizational behavior (DiMaggio & Powell, 1983; Scott, 2008). In the case of emerging economies, PFM systems are in many cases heavily influenced by bureaucracy, hierarchy, and incentives driven by compliance that delay innovation and responsiveness (Allen & Tommasi, 2001). *In this institutional view, agile governance in PFM consists of:*

-Institutional alignment: The legislative, regulatory, and procedural framework should allow flexibility when budgeting, procurement, and treasury operations (IMF, 2020).

-Organizational legitimacy: Fate favors initiatives that demonstrate agile approaches as credible in the governance norm of stakeholders like legislators, civil servants, auditors, and citizens (OECD, 2021).

- Path dependency awareness: Awareness of past institutional constraints helps design phased, context-sensitive agile interventions that do not contravene legal or procedural mandates (Cangiano et al., 2013). Institutional Theory states that successful reforms of agile PFM should be aware of navigating formal rules and

informal practices which would then allow slowness in cultural and procedural transformation.

Public Value Governance

In public administration, unlike other areas, value is more than just legal compliance or internal efficiency. The first step to bringing fiscal control towards being more focused on social impact relating to citizen welfare is by shaking off the guilders of PFM from pure technicist fiscal control:

- **Citizen-centric budgeting:** Agile PFM allows deciders to reallocate funds after receiving citizen feedback, social results, or changing priorities (OECD, 2020).
- **Transparency and accountability:** With iterative reports, real-time dashboards, and the use of open data, trust is built, and the social impacts of fiscal decisions are proven (World Bank, 2019).
- **Collaborative governance:** Agility in approach, teams working across functional boundaries and stakeholder engagement allow for co-creation of policies and interventions right through to public value creation (Mergel et al., 2021). Thereby a better assurance that the running of public value governance through PFM had no threats to accountability and social objectives given the agile frameworks.

Integrating Theories for Agile PFM

By blending Complexity Theory, Institutional Theory, and value governance, an extremely potent framework emerges to be effective in analyzing and designing agile public finance management systems for late starters:

Theory	Contribution to Agile PFM
Complexity Theory	Supports iterative planning, adaptive decision-making, and feedback-driven fiscal responsiveness.
Institutional Theory	Guides reform within existing formal and informal rules, ensuring legitimacy and sustainability.
Public Value Governance	Ensures that agile practices align with societal goals, transparency, and citizen-centered accountability.

This is the integrated theoretical framework that provides the basis for the proposed Agile PFM Model by this study. The model explains why there is a requirement that governments have adaptive structures, legal flexibility, digital platforms, and cross-functional collaboration to be both efficient and public value-compliant in fiscal management and thus provides a lens to evaluate PFM reforms in Nigeria, Ghana, Kenya, India, the United States, and the United Kingdom in country-specific contexts.

Agile Governance Principles for Public Financial Management

The very heart of agile governance is to provide a framework whereby systems of Public Financial Management (PFM) would become adaptive, transparent, and efficient. Built on principles of agile software development, agile governance means iterative planning with fast feedback loops, teamwork, and learning (Mergel et al. 2021; Luna et al. 2020). Within PFM, these principles can enhance budget planning, procurement, treasury functions, and auditing, making them both flexible and responsive, yet accountable.

Iterative and Adaptive Budgeting

Traditional Public Financial Management (PFM) processes usually constrain themselves to an annual, historical allocation-based static forecasting. Introduced by agile governance is iterative budgeting that continually adjusts and updates the fiscal plans based on real-time data, stakeholder feedback, and performance outcomes (OECD, 2020). For example, in Kenya, the real-time expenditure dashboards together with IFMIS facilitate mid-year budget reallocations in favor of emergency needs arising from drought or health crises (World Bank 2020). Moreover, Governments could institute rolling forecasts, quarterly budget reviews, and scenario-based planning, thereby allowing greater fiscal flexibility in the accurate allocation of such funds (IMF 2020). Iterative budgeting carries forward the tenets of Complexity Theory in understanding that public finance is dynamic and that decisions must change with changing circumstances (Holland 1995; Snowden & Boone 2007).

Cross-Functional PFM Teams

Agile governance invokes an inter-departmental enterprise culture and interdisciplinary involvement in public finance processes stewardship by fostering cross-functional collaboration (Mergel et al., 2021). Coordination in cross-functional teams across various disciplines then reduces time lost in bureaucratic processes and improves problem-solving. A typical example will be India's PFMS, which combines finance, procurement, and sectoral program teams for effective oversight of fund disbursement, bottleneck detection, and timely delivery of social programs (Mukherjee & Sikdar, 2020). Additionally, establishing agile PFM teams will enable responding and coordinating budgeting, auditing, procurement, and treasury activities while enriching institutional learning, thus reducing delays in funds allocation (OECD, 2021). That is also backed by this principle, which falls under Institutional Theory, which accepts that performing a task involves some balance between the formal rules surrounding such endeavors and collaborative behavior in the bureaucratic space built into these endeavors (DiMaggio & Powell, 1983; Scott, 2008).

Rapid Feedback Loops and Performance Monitoring

The agile public financial management system is a continuous monitoring and feedback mechanism that identifies problems, informs better spending decisions, and enhances accountability. Feedback loops enable governments to detect early in the fiscal cycle detect instances of poor performance, misuse of funds, or deviations from policy objectives (Long et al., 2023) that manifest in their fiscal systems. According to Adewuyi (2022), an integrated real-time expenditure data of TSA online platforms in Nigeria may assist politicians and auditors in knowing what performance areas require intervention. Agile PFM applications now include dashboards, key performance indicators, and automated alerts to enable one decision iteration for always aligning fiscal and strategic agendas (OECD, 2020). On another note, real-time feedback systems can

also serve Public Value Governance by linking financial decisions to measurable social outcomes and citizen satisfaction (Bryson et al), 2014).

Experimentation and Regulatory Sandboxes

By flourishing interventions regarding public finances management, it continues facilitating governments to try out new modalities in public finance management at a very low risk prior to adopting them as nationwide policies (OECD, 2024). Mildly, this was constituted throughout the agile pilot procurement processes through the Government Digital Service (GDS) specifically commissioned in the United Kingdom to provide positive improvements towards the procurement process (UK Cabinet Office, 2019).

In emerging countries, this scenario naturally provides enabling legislation for testing the effectiveness of digital acquisition apparatus, performance-based budgeting innovations, or new auditing approaches before replication at the practical level (Mergel et al., 2021). Hence, experiments support complexities by way of decision-making on an open format that would allow governments to confront emergent challenges without committing disproportionate resources or breaking existing institutional conventions.

Data-Driven Decision-Making

Indeed, the data-oriented approaches for agile public financial management are in evidence-based fiscal decision making as real time analytics, predictive models and digital dashboards often used to leverage efficiency, openness and discipline in fiscal management (Long et al., 2023; OECD, 2021). For example, India, in her PFMS, and the United States Digital Service are two of many countries that make extensive use of data analytics aimed at predicting expenditures, tracking the usage of funds and even detecting violations in the financial operations of the country (Mukherjee & Sikdar, 2020; GAO, 2021). This is where governments can utilize data in deciding on incremental budget modifications or towards identifying high-priority programs and enhancing accountability for public funds under both Complexity Theory and Public Value Governance (Moore, 1995; Snowden & Boone, 2007).

Integration of Agile Principles in PFM Processes

The synergistic implementation of these principles- iterative budgeting, cross-functional teams, rapid feedback, experimentation, and data-driven-decisions- is the foundation for Agile PFM. The PFM model enhances fiscal resilience, improves service delivery, and deepens accountability by:

1. Flexibly catering resource allocation to changing needs.
2. Alleviating silo barriers and improving inter-department coordination.
3. Ensuring real-time monitoring and immediate corrective action.
4. Providing space for innovation and experimentation in fiscal processes.

By incorporation of the above principles, emerging economies can narrow the gap between traditional very rigid PFM systems to the dynamic requirements of modern public finance (OECD, 2020; Mergel et al. 2021).

Country-Specific Applications of Agile Governance in Public Financial Management

Models of agile governance under Public Financial Management (PFM) differ from country to country because of the peculiarities of institutional capacity and its digital infrastructure, political commitment, and

background history of previous rounds of reform measures. In a comparative analysis, it was able to showcase the degree to which some nations-Nigeria, Ghana, Kenya, India, USA, and UK, are embedding agile principles within their environment (i.e. budgeting, procurement, treasury, and financial oversight). These cases provide concrete instances of how agile governance could be used in greater transparency, thereby reducing duplication in work and cost-efforts toward fiscal resilience.

Nigeria

Since October 2023, Nigeria's Public Finance Management reforms have been making strides towards further transparency and better cash management, a development that would not have been possible without the ongoing digital transformation efforts. The Treasury Single Account (TSA) has served as one of the most important reforms that eliminated thousands of fragmented government accounts and unified the cash flows of the government under one platform. Alongside the TSA initiatives Nigeria has been building on e-procurement efforts, as well as enhancing the Integrated Payroll and Personnel Information System (IPPIS) that is purposed to eliminate ghost workers and strengthen expenditure management (Oloruntoba, 2021).

Such reforms give rise to partnerships among the finance authorities, IT units, and anti-corruption bodies- cross-functional teamwork to be seen in agile practices. However, overall, there are quite a few challenges. The building of limited institutional capacity, political interference, and shifts of agenda tend to be barriers to the total embrace of agile governance among the public finance management system. For sustenance, Nigeria needs to put more priority towards strengthening data governance systems and enhancing digital literacy within the civil service (World Bank, 2020).

Ghana

With the introduction of the Ghana Integrated Financial Management Information System (GIFMIS), Ghana's Public Financial Management (PFM) scene is changing into new dimensions of modernization in budgeting, accounting, and procurement processes (World Bank, 2022). GIFMIS, referring to agile principles in respect of its modular implementation, allows different modules to be digitized in stages. Examples of modules include budget preparation, expenditure controls, and cash management. Ghana has also established a medium-term expenditure framework (MTEF) tied to performance-based budgeting, which allows it to engage in relatively flexible fiscal planning (IMF 2022). Very much agile governance is embodied in government efforts toward real-time dashboards on tracking expenditures in social sectors, such that policymakers would be able to adjust with respect to emerging priorities. There are currently challenges regarding interoperability between government databases and real-time data accuracy. Nevertheless, Ghana's commitment to digital governance positions it as a prime example of agile PFM uptake in West Africa.

Kenya

The name of Kenya has become synonymous with successful digital transformation in public administration. The Integrated Financial Management Information System (IFMIS) and the eCitizen platform bear adequate evidence for the adoption of agile principles-in aspects of iterative system upgrades, user-centered design, and real-time financial data availability (World Bank, 2020). The migration to e-procurement (IFMIS e-procurement module) has greatly reduced delays in procurements and increased audit traceability for visibility and early detection of issues, thereby adhering to agile practices. Supporting the use of mobile

payments for government transactions is yet another way Kenya promotes agile principles by digital ecosystem integration to enhance service delivery.

Agile PFM has been invaluable during crisis times, such as during the COVID-19 pandemic, when digital dashboards in Kenya enabled rapid reallocation of health funds and expedited procurement of emergency supplies (OECD, 2021). Threats include cybersecurity risks, legacies at county-level systems, and sporadic resistance to change by a section of public officials.

India

It is one such illustration of agile PFM in the large one-on-one emerging economies. The public financial management system (PFMS) holds the capability of tracking in real-time disbursement of funds from the Central and State grants, which in turn supports the iterative monitoring and the rapid corrective action for central and state government programs (Mukherjee & Sikdar, 2020).

Pilot testing, phased rollouts, and evidence-based scaling all express the agile principles embodied within India's Aadhaar-linked Direct Benefit Transfer (DBT) system. It has reduced leakages and added a sharper targeting mechanism while increasing transparency (UNDP, 2019).

Implementation of Government e-Marketplace (GeM) embodies the principles of agile procurement including the notions of transparency, competition, and iterative improvements based on feedback from vendors and buyers (Government of India, 2021). India's national reforms showcase the advantages associated with inter-ministerial collaboration complemented by state governments and digital teams. Conspicuous challenges include parochial disparities in the states, concerns regarding data privacy, and infrastructure gaps in the peripheries.

United States

Broadly speaking, agile governance in the U.S. relates to federal digital transformation initiatives rather than comprehensive public financial management (PFM) reform. Initiatives like the U.S. Digital Service (USDS) and 18F propagate agile methodologies for federal IT procurement and service delivery (GAO, 2021). Iterative budget reforms have been undertaken in the U.S., including performance-based budgeting and evidence-based policymaking frameworks, which tend to enhance responsiveness and accountability (OMB, 2020). For instance, the DATA Act initiative intends to standardize means of financial data across federal agencies so that real-time analytics and data-driven decision-making can be exercised-real agile principles (Treasury & OMB, 2019). The United States enjoys PFM where agile governance takes place with cross-agency councils and data-driven oversight mechanisms, while obstacles are the bureaucratic resistance, complexities with procurement, and vulnerabilities of cyberspace.

United Kingdom

Agile public governance is, as expected, very much a by-product of the UK. Amongst such initiatives is the Government Digital Service (GDS) that sets out agile delivery benchmarks around user-centered design, cross-functional teams, and iterative development cycles (UK Cabinet Office, 2019). To illustrate, the Whole of Government Accounts (WGA) program in the UK is meant to improve fiscal transparency such that it should make agile budgeting operational through the integration of real-time finance data from the agencies (NAO, 2020). Digital procurement reform through supplier engagement sprints and sandboxed procurement test-run

is an illustration of agile principle experimentation and quick learning (OECD, 2024). Accompanying reforms in the public finance of the UK are also predictive analytics for fiscal forecasting and fraud detection to enable evidence-based adaptive management. However, a vast array of challenges arises in trying to balance the competing forces of regulatory complexity and transparency and to make the various legacies interoperable.

Comparative Insights Across the Six Countries

Country	Agile Strengths	Main Barriers
Nigeria	TSA, e-procurement, real-time visibility	Political interference, low digital capacity
Ghana	GIFMIS modular rollout, MTEF	Data accuracy, interoperability
Kenya	IFMIS, mobile payments, dashboards	Cybersecurity, local system disparities
India	PFMS, DBT, agile procurement (GeM)	Privacy concerns, state-level capacity gaps
United States	USDS, DATA Act, iterative budgeting	Bureaucratic inertia, procurement complexity
United Kingdom	GDS leadership, WGA, experimentation	Regulatory constraints, legacy systems

Implementation Roadmap for Agile Public Financial Management (PFM)

The mold of PFM into agility thus requires a conscious and deliberate unlearning of all erstwhile practices-atmosphere of sorts of adaptationism, evidence-orientation, and collaboration. For many emerging economies, this mix of changes is concerted action for institutional reform, digital modernization, and sustained capacity development in the same breath. There follows a six-headed strategic phase roadmap for governments to transition towards agile fiscal governance.

Phase One: Diagnostic Assessment

Thus begins the transformation-the study of the current public financial management environment: its administrative processes and technology capabilities, as well as established institutional constraints. This extensive assessment typically includes budget formulation and execution workflows, cash management practices, procurement mechanisms, audit procedures, and interagency coordination. Standardly established tools-such as those named after Public Expenditure and Financial Accountability (PEFA), IMF diagnostic instruments, and World Bank fiduciary assessments-help pinhole existing gaps in transparency, integration within systems, and timeliness. In case of good execution, this diagnostic phase can allow decision-makers to find priority areas where rapid methods would result in immediate, dramatic change, such as procurement processes, budget monitoring, and expenditure audits.

Phase Two: Stakeholder Alignment and Governance Structure

After understanding the diagnostic insights, the next step is to put in place definite governance and accountability

mechanisms. Normally, this phase includes the forming of:

- Cross-functional Agile PFM Teams that take budget analysts, ICT specialists, procurement officers, auditors, and policy planners together.
- Steering Committees are designated to be responsible for political guiding and high-level coordination across ministries.
- Change Management Plans are meant for building buy-in, reducing resistance, and preparing staff for new workflow.

Most often in the developing context, reform seems to become stagnant: in many cases due to burnout, lack of ownership, or change in political priorities. It is, therefore, essential to engage stakeholders early to build a consensus around a common vision and ensure an institutional commitment during the entire life cycle of the reform.

Phase Three: Agile Policy and Process Redesign

Currently, public finance management processes are being converted to principles of agility, such as incremental rollout, continuous feedback, and adaptive planning. Key activities include:

- Shortening the long budget cycle into short, iterative periods for review.
 - Prepare Minimum Viable Process (MVP) in procurement, disbursement, or expenditure tracking functionalities.
 - Create policy sandboxes for limited and controlled experimentation, for possible future national adoption.
 - Integrated real-time transparency, such as digital dashboards and automated audit trails.
- Moving from rigid yearly cycles to flexible, iterative processes will allow governments to be ever more capable of rapid response to emergent fiscal pressures, shocks in the economy, or challenges in implementation.

Phase Four: Digital System Modernization

The major features in the process of digital modernization become the technological landscape that will cater for agile public financial management. The new systems are designed for automation and interoperability and to provide real-time analytics for faster and more accurate decision-making in fiscal matters. This phase shall cover:

- Upgrading legacy IFMIS from a platform of non-modular design to a modular design with API-enabled systems.
- Building a Central PFM Data Layer that brings together information about budgets, revenues, expenditures, and procurements for a single analytical environment.
- Leveraging advanced technologies such as cloud computing, open data portals, and automated workflow engines.

These enhancements truly make the discounted agile practice more scalable and reduce manual flow-dependent delays and potential touch points for failure.

Phase Five: Pilot Implementation and Iterative Refinement

The purpose of pilot initiatives is to test agile methodologies in a controlled environment prior to implementation at scale. The government may select certain ministries or sectors to pilot innovations, such as:

- Iterative budgeting in health or education ministries.
- Agile procurement models within infrastructure or public works agencies.
- Real-time expenditure monitoring dashboards in finance ministries. They work in short cycles, typically lasting for periods of 30-60 days, with defined goals, continued input from stakeholders, and a high degree of reactivity to change. The knowledge gained from these pilots will facilitate system-wide design changes, digital tool updates, and adjustments in the training program.

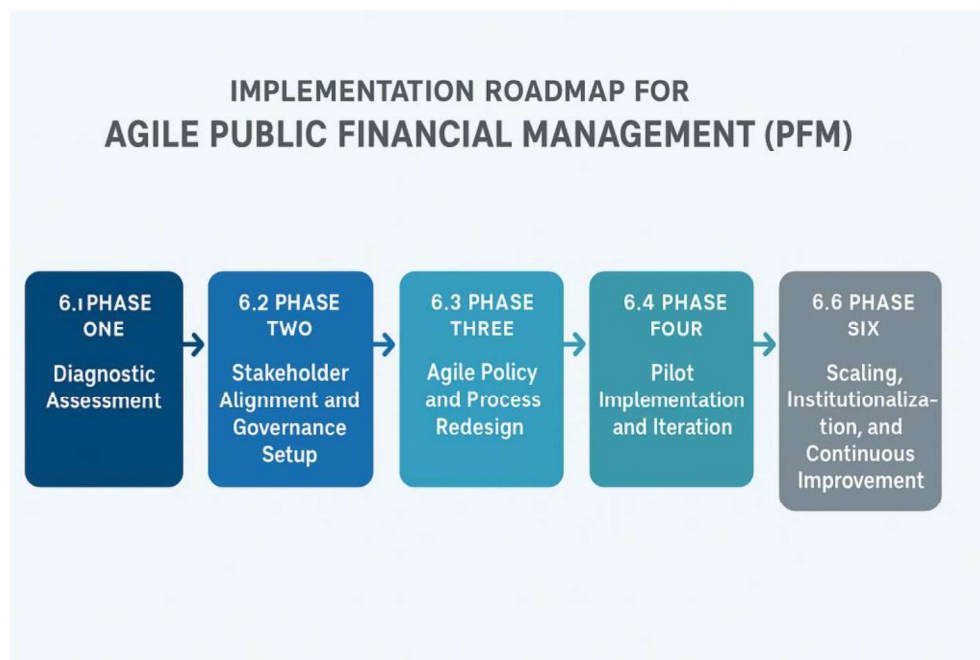
Phase Six: Scaling, Institutionalization, and Continuous Improvement

Embed agile PFM practices in the national systems at final phase to ensure continuity over time, such as:

- Revising all policy, regulation, and operational guidelines that align with the formalizing of agile practices.
 - Training for PFM staff on agile methodology, data, and digital tools, which will be an ongoing professional development program.
 - Establishment of cross-ministerial digital platforms and dashboards for use by subnational units.
- Institutionalization reinforces sustainability while providing less potential risk to revert to the previous legacy. Continuous improvement, based on real-time data and performance metrics, ensures the responsiveness of PFM systems to the changing political economy and technology.

Figure 1: Implementation Roadmap for Agile Public Financial Management (PFM)

This figure shows the sixfold strategic road map to Agile Public Financial Management system integration. In the figure is the journey of transformation that starts from the diagnostic assessment to take up stakeholder alignment, agile policy re-design, back to the modernization of digital systems, piloting, and finally national scale institutionalization. The model shows the interdependent nature of governance reform, iterative feedback loops, and data-driven decision processes that build adaptive, transparent, and resilient PFM environments in emerging economies.



Policy Recommendations

Generally, when it comes to many of the powerful transformation initiatives for Public Financial Management (PFM) systems in emerging economies, Agile governance provides the steering mechanism. Successful implementation requires very well-calibrated policy interventions considering the above-mentioned constraints vis-a-vis institutional arrangements, digital capital, and human capital. Below are comprehensive sets of policy recommendations, backed by international evidence, and in concordance with the study's findings.

Strengthening Legal and Regulatory Foundations for Agile PFM

The acceptance of agile public finance management (PFM) in emerging economies is today greatly hampered by the traditional, rigid legal framework provided for budgeting, procurement, and auditing. In doing so, countries should thus be updating antiquated acts on public finances, procurement laws, audit regulations, and financial procedures to allow iterations, swift changes, and digital approvals.

Agile reforms are said to succeed only with beneficial laws in support (OECD, 2020). The United Kingdom, for example, grants flexibility to budget adjustments mid-year in its Public Finance Management Act, with digital signatures affording agile adoption of this PFM by all governmental departments (GDS, 2022). Similarly, India's own transition to the Public Financial Management System (PFMS) needed the redrafting of substantial legal provisions to allow for real-time digital payments and automated budget releases (World Bank, 2022).

Legal reforms must be made:

- Iterative budget cycles instead of rigid annual plans.
- Digital verification and e-approvals of financial transactions.
- Real-time fiscal reporting requirements for transparency.

-Agile procurement models, including framework agreements and rapid bidding.

The delay in the rehabilitation of obsolete laws on procurement and PFM in Nigeria and Kenya is hindering digital transformation (IMF, 2023). In other words, modernized legal frameworks are key to enabling scalability of agile PFM.

Institutionalize Cross-Functional Agile PFM Teams

Agile governance calls for collaborating and coordinating actions across professions. When these countries are concerned, institutionalization of the multidisciplinary Agile PFM Teams is necessary, including budget, procurement, auditing, IT, treasury, and policy professionals.

Empirical literature shows that cross-functional teams speed up decision-making and eliminate fiscal operations bottlenecks (Mergel et al., 2021). A few examples:

-Ghana's GIFMIS reform team included IT specialists along with treasury and procurement officers and was instrumental in fast-tracking IFMIS deployment (World Bank, 2019).

- In Kenya, the e-procurement team adopted the Agile methodology at one time to allow simultaneous testing of the system across multiple agencies so they could learn from one another (CABRI, 2021).

-The U.S. Office of Management and Budget (OMB) utilize cross-functional digital teams to shorten IT procurement timelines and enhance transparency (OMB, 2020). Therefore, the Agile PFM Teams must also be sustained by embedding them into Ministries of Finance permanently instead of just remaining as temporary reform committees.

Expand Digital Infrastructure and Data Interoperability

Digital modernization is at the center of agile public financial management (PFM), with all the components of the reality that digital modernization brings about increasingly becoming automation, transparency, and real-time fiscal decision-making towards PFM. Thus, modernization at all levels of IFMIS and other related platforms becomes an absolute must for the PFM system modernization. International examinations reveal that countries with interoperable digital public financial management architectural systems perform better in budget execution and leakages (IMF, 2022; OECD, 2021). Some of the most important reforms include:

-Moving from monolithic IFMIS systems to modularized IFMIS systems such as in India, Ghana, and the U.K.

-Mechanisms for API-enabled data sharing with tax authorities, procurement agencies, and audit institutions so that one gets a holistic view of public expenditure.

-Real-time dashboards for treasury operations, procurement tracking, and revenues analytics.

The benefits of automation are aptly demonstrated by the Nigerian Treasury Single Account (TSA); after total digitalization, recorded federal revenue leakages have dropped sharply (IMF, 2021). The IFMIS e-procurement module in Kenya has cut down procurement cycle times and has made an increase in audit trialability (CABRI, 2020). Therefore, a government is assured of a digital investment without the prospective security exposure in creating an updated law on the data protection itself.

Implement Pilot Programs Before National Rollout

Testing by a small group to determine feasibility of function is called pilot testing: with pilot authorities, low-level programs launch and from that point learn, accommodate, and ramp up. Empirical evidence shows that pilot programs reduce the risks involved in reforms, thereby facilitating user acceptance (World Bank, 2020).

Examples of pilot options are:

- Agile budgeting in one or two ministries.**
- Digital procurement sandboxes.**
- Automated audit trail pilots.**
- Real-time expenditure dashboards.**

These are examples:

-PFMS was piloted among certain states in India before nationwide rollout, which helped reduce the flaws in design by 30% (World Bank, 2022).

-The e-procurement pilot in Ghana was meant to streamline vendor onboarding processes before the full-scale rollout to all ministries (GIFMIS Secretariat, 2019).

-The Agile Delivery Model in the UK emphasizes iterating pilots before scaling the digital services that users have. (GDS, 2022). In this program, structured sprint cycles between 30 and 90 days will be employed, depending on evaluation briefs to be established at the end of each sprint.

Strengthening Capacity Building for PFM Officers

Agile PFM introduces new requirements for competencies in digital tools, financial analytics, automation, and agile methods. Any capacity development effort must therefore be continuous and layered. Research shows that the main impediment to PFM modernization is glaringly capacity constraints, not technology (Allen, Hemmings & Potter, 2019; IMF, 2023). Recommended actions are:

- National PFM academies or expert "Centres of Excellence" should be established.**
- To partner with such institutions as CABRI, IMF AFRITAC, and World Bank GovTech programs;**
- Training should be provided in:**
 - Agile project management.**
 - Data analytics and visualization.**
 - Digital procurement systems.**
 - Treasury automation.**

By linking training attendance to promotion in the civil service, the Ministry of Finance in Kenya improved the adoption of IFMIS as a training regime (National Treasury of Kenya, 2021). Nigeria and Ghana have undertaken similar reforms.

Enhance Citizen Engagement and Fiscal Transparency

Governments would be wise to implement:

- An open budget portal for citizens monitoring such spending.**
- A feedback mechanism such as mobile platforms for reporting public procurement irregularities.**
- Participatory budgeting successfully applied in India and in select counties in Kenya.**

The “Open Fiscal Data Dashboard” in Ghana improved civil society engagement considerably (World Bank, 2021). These provide global benchmarks for real-time accountability and transparency: USAspending.gov, in the U.S., and Open Treasury Data, in the U.K.

Promote Regional and International Collaboration

Cross-country learning accelerates PFM reforms in their stride. Regional organizations such as ECOWAS, AfDB, CABRI, and IMF AFRITAC offer opportunities for policy exchange and peer learning.

-Ghana and Nigeria were peer reviewed and shared their procurement reforms (CABRI, 2022).

-Kenya adopted aspects of India's PFMS architecture to automate payments (World Bank, 2022).

-The U.K. and U.S. provide best practices globally in agile delivery of digital government (GDS, 2022; OMB, 2020).

Members of an organization, particularly individuals involved in the venture capitalist start-ups, need comprehensive instructions concerning standard protocol for their representation concerning the venture capitalist.

Ensure Sustainability Through Monitoring and Evaluation (M&E)

Robust reporting mechanisms, a scrutiny of programs, projects, policies, and operations, as well as concrete feedback on achievements and impacts are key aspects of sustainability. Research has shown that these reforms have not always succeeded where the M and E system is either weak or does not exist (IMF 2020).

Some interventions that the government should adopt include:

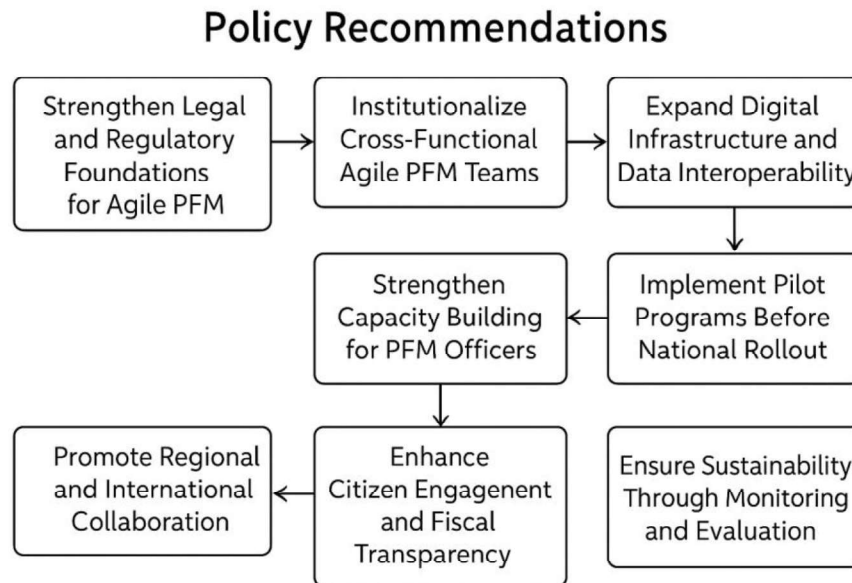
-Key Performance Indicators (KPIs) - these are needed for transparency, efficiency in spending, the time taken for procuring and audit compliance.

-Budget execution performance and procurement performance quarterly reviews.

-Annual evaluation of agile maturity like the ones conducted in the UK and the US.

Automating M&E dashboards has helped put a significant rein on highlighting delay flags toward government payments and curbing inefficiencies in India, according to the PFMS (World Bank, 2022).

Figure 2. Policy Recommendations for Strengthening Agile Public Financial Management (PFM) Systems



It is illustrated in this figure as a set of systematic policy actions aimed at embedding agility within national Public Financial Management (PFM) systems. From foundational reforms such as strengthening legal frameworks and the establishment of cross-functional teams for PFM-the recommendations move to more advanced steps of digital modernization, skills building, and the trial of agile processes before nationwide adoption. The model also includes citizen engagement and fiscal openness, which together with monitoring and evaluation provide continuity over the long term. Regional and international collaboration as networks reinforced through collective learning between governments is positioned as a supporting pillar enhancing the exchange of knowledge and interoperability.

Conclusion

Adoption of Agile Public Financial Management (PFM) practices stands out among present-day options a government may choose to achieve a transparent-based fiscal governance with an open and responsive approach to sustainability. Evidence indicated that traditional PFM systems in Nigeria, Ghana, Kenya, India, the U.S., and U.K. have encountered numerous complications arising from the manual procedures, standalone data, and slow reporting processes that can no longer satisfy contemporary governance requirements (Allen, Hemming, & Potter, 2013; Andrews, Pritchett, & Woolcock, 2017). Agile PFM responds to these needs by emphasizing iterative reforms, continuous feedback, modular digital solutions, and enhanced stakeholder engagement. The agile approach to public financial management enhances reporting, improves forecasting of receipts, and makes for better accountability with a reduction in fiscal leakages (OECD, 2021; World Bank, 2022). India's digital public infrastructure and Kenya's Treasury Single Account testify to better public finance performance within an agile-digital-linkage framework.

One majorly limiting factor is the extent to which digital capacity is present, and these limitations have

augmented certain areas, thus making the adoption even slower with political acceptance, cybersecurity, etc. (Dener, Watkins, & Dorotinsky, 2011). The roadmap for implementation, accompanied by the recommendations for policies put forth in this study, acknowledges the phased-in reforms, need for political sponsorship, regulatory modernization, and orientation of the workforce in breaking such barriers.

Overall, Agile PFM provides a clear, practical model for governments to modernize financial management systems and build trust with citizens. As digital transformation accelerates globally, adopting agile principles will be essential for achieving fiscal resilience and delivering efficient, transparent public services.

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